

DATA MINING L

Submission Report

KSE-100 Pakistan Stock Exchange | 2019 2024

Dataset	KSE100_5Years_2019_2024.csv
Language	Python 3 (pandas, numpy, matplotlib, seaborn, scipy, plotly, streamlit)
Records	148,863 raw rows → 133,248 after cleaning

LAB #07 Data Cleaning & Exploratory Data Analysis

A1. Project & Dataset Description

Project Title: Pakistan Stock Exchange (KSE-100) Historical Performance Analysis

Problem Statement: Analyse five years (2019-2024) of daily OHLCV trading data for KSE-100 listed companies to identify price trends, volatility patterns, volume behaviour, and relative stock performance.

Dataset: KSE100_5Years_2019_2024.csv a daily price feed covering all major KSE-100 listed equities. Each row represents one trading day for one company and includes the opening price, intraday high and low, closing price, and total shares traded.

A2. Shape & Structure

Attribute	Value
Rows	148,863
Columns	8
Date Range	2019-01-01 to 2024-12-31
Companies	Multiple KSE-100 listed symbols
Numerical Features	OPEN, HIGH, LOW, CLOSE, VOLUME
Categorical Features	DATE, SYMBOL, COMPANY

A3. Descriptive Statistics Summary

From df.describe(), key observations:

- Close price ranges from PKR 1.57 to PKR 8,065.34, indicating a wide spread across small-cap and large-cap stocks.
- Mean close price is PKR 230.62 with a high standard deviation of 614.14, confirming significant price dispersion.
- Volume ranges from 0 to 481 million shares per day; the median (155,600) is far below the mean (1.6M), indicating a right-skewed distribution driven by a few high-volume trading sessions.

B1. Missing Values

Detection: df.isnull().sum() was applied to all 8 columns.

Result: No missing values were found in any column (all counts = 0).

Justification: The dataset was sourced from a structured exchange feed with pre-validated OHLCV entries. No imputation was required.

B2. Duplicate Records

Detection: df.duplicated().sum() returned 0 duplicate rows.

Result: No rows were removed. The dataset contained unique (date, symbol) combinations throughout.

B3. Data Type Formatting

- DATE column was converted from string (object) to datetime64 using `pd.to_datetime()`. This enables time-series operations such as resampling and year/month extraction.
- SYMBOL column was converted to category dtype to reduce memory usage and improve groupby performance.

B4. Zero-Volume Rows

15,615 rows had VOLUME = 0, representing trading days where no shares were exchanged (holidays, circuit breakers, or data gaps).

Decision: These rows were removed as they do not represent real market activity and would distort volume-based analyses.

Remaining rows after removal: 133,248.

B5. Outlier Detection IQR Method (VOLUME)

IQR analysis on the VOLUME column:

- Q1 = 19,500 | Q3 = 1,297,576 | IQR = 1,278,076
- Lower bound = -1,897,614 | Upper bound = 3,214,690
- Outliers detected: 16,998 rows with volume > 3.2M

Decision: Retained. Extreme volumes correspond to real high-activity trading days (earnings releases, index rebalancing, market rallies). Removing them would lose economically meaningful data.

B6. Outlier Detection Z-Score Method (CLOSE)

Z-score analysis on the CLOSE column identified 1,472 rows with $|z| > 3$.

Decision: Retained. These correspond to genuinely high-priced stocks (e.g., blue-chip companies with share prices above PKR 2,000). They are valid market observations, not data errors.

B7. Feature Engineering

Four new features were derived to enrich the dataset for EDA:

- Daily_Return (%): Percentage change in CLOSE price per stock per day computed using `groupby(SYMBOL).pct_change()`. Captures day-to-day return behaviour.
- Price_Range: Intraday range = HIGH – LOW. Measures daily price volatility.
- Year: Extracted from DATE for annual trend analysis.
- Month: Extracted from DATE for seasonal pattern detection.

Final cleaned dataset shape: 133,248 rows × 12 columns. Saved as `lab07_outputs/psx_cleaned.csv`.

Challenges Faced

- The DATE column was stored as a plain string, requiring explicit conversion before any time-series operations could be performed.
- The heavy right-skew in both price and volume required log transformation to make distributions visually interpretable.
- Zero-volume records were ambiguous (could represent holidays or data errors); the decision to remove them was made based on their distortion of volume statistics.

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(env) PS D:\data_mining> & d:\data_mining\env\Scripts\python.exe d:/data_mining/lab7.py  
=====
```

PART A - DATASET UNDERSTANDING

[A1] First 5 records:

	DATE	SYMBOL	COMPANY	OPEN	HIGH	LOW	CLOSE	VOLUME
0	2019-01-01	ABL	Allied Bank Ltd.	45.415128	45.415128	44.454583	45.242397	77400
1	2019-01-02	ABL	Allied Bank Ltd.	44.488285	45.027537	44.066995	44.235512	298100
2	2019-01-03	ABL	Allied Bank Ltd.	44.656796	45.288732	44.652582	44.875866	550300
3	2019-01-04	ABL	Allied Bank Ltd.	44.530412	45.288736	44.530412	45.078091	111800
4	2019-01-07	ABL	Allied Bank Ltd.	45.078090	47.142414	45.078090	46.940193	1652400

[A2] Rows : 148,863

Columns : 8

Names : ['DATE', 'SYMBOL', 'COMPANY', 'OPEN', 'HIGH', 'LOW', 'CLOSE', 'VOLUME']

[A3] Data Types:

DATE	str
SYMBOL	str
COMPANY	str
OPEN	float64
HIGH	float64
LOW	float64
CLOSE	float64
VOLUME	int64

[A4] Numerical : ['OPEN', 'HIGH', 'LOW', 'CLOSE', 'VOLUME']

Categorical : ['DATE', 'SYMBOL', 'COMPANY']

[A5] Descriptive Statistics:

	OPEN	HIGH	LOW	CLOSE	VOLUME
count	148863.00	148863.00	148863.00	148863.00	1.488630e+05
mean	230.57	233.14	228.10	230.62	1.617878e+06
std	613.83	618.82	609.20	614.14	5.965506e+06
min	1.59	1.59	1.55	1.57	0.000000e+00
25%	22.49	22.80	22.11	22.44	5.500000e+03
50%	71.19	72.02	70.21	71.07	1.560000e+05
75%	188.13	190.60	186.00	188.37	1.079823e+06

```
=====
DATASET: KSE-100 Pakistan Stock Exchange Historical Data
PERIOD : 2019 - 2024 (5 Years)
CONTENT: Daily OHLCV data for KSE-100 listed companies
GOAL   : Analyse stock price trends, volatility, volume
        patterns, and sector performance over 5 years.
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PART B - DATA CLEANING
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```
Detected columns : {'SYMBOL': 'SYMBOL', 'COMPANY': 'COMPANY', 'OPEN': 'OPEN', 'HIGH': 'HIGH', 'LOW': 'LOW', 'CLOSE': 'CLOSE', 'VOLUME': 'VOLUME'}
Date column      : DATE
```

```
[B1] Missing Values per Column:
```

```
DATE      0
SYMBOL     0
COMPANY    0
OPEN       0
HIGH       0
LOW        0
CLOSE      0
VOLUME     0
```

```
✅ No missing values detected.
```

```
[B2] Duplicate rows: 0
     Rows after removal: 148,863
```

```
[B3] 'DATE' → datetime ✅
     'SYMBOL' → category ✅
```

```
[B4] Zero-volume rows: 15615
     Removed. Remaining rows: 133,248
```

```
[B5] IQR Outliers in 'VOLUME':
     Q1=19,500 Q3=1,297,576 IQR=1,278,076
     Bounds=[-1,897,614, 3,214,690] Outliers=16,998
     → Retained (extreme volumes are real market events)
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```
[B5] IQR Outliers in 'VOLUME':
     Q1=19,500 Q3=1,297,576 IQR=1,278,076
     Bounds=[-1,897,614, 3,214,690] Outliers=16,998
     → Retained (extreme volumes are real market events)
```

```
[B6] Z-score Outliers in 'CLOSE' ( $|z| > 3$ ): 1,472
     → Retained as valid extreme prices
```

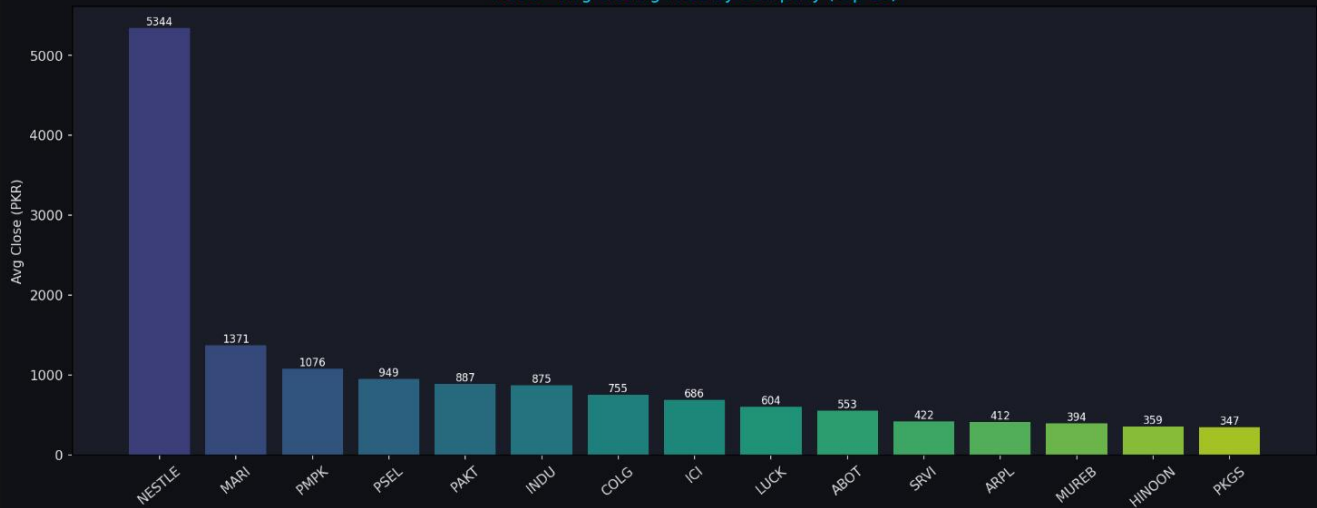
```
[B7] New features: Daily_Return, Price_Range, Year, Month ✅
```

```
✅ Cleaned dataset shape: (133248, 12)
```

Plot 3 - Close Price Trend (Top 5 Stocks)



Plot 5 - Avg Closing Price by Company (Top 15)

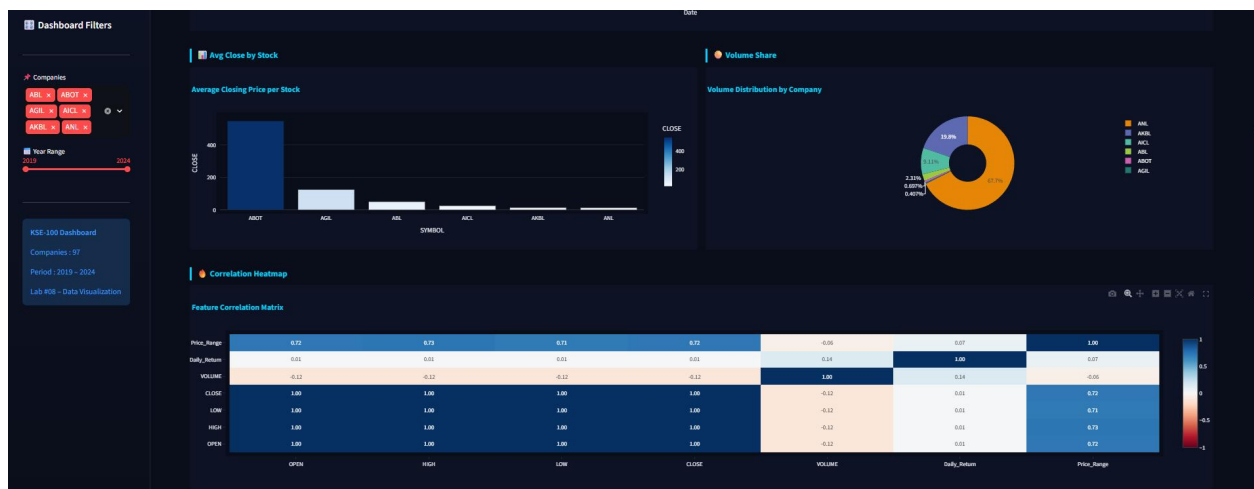


LAB #08 Data Visualization Dashboard

Part A Project Overview

Project Title	KSE-100 Pakistan Stock Exchange Interactive Analytics Dashboard
Problem Statement	Provide a unified, interactive visual interface for exploring 5 years of KSE-100 trading data, enabling trend analysis, risk assessment, and comparative stock evaluation without requiring programming knowledge.
Dataset	KSE100_5Years_2019_2024.csv 148,863 daily OHLCV records for KSE-100 listed companies (2019-2024).
Dashboard Purpose	Transform cleaned stock data into interactive charts and KPI cards that allow stakeholders to monitor market performance, compare companies, and identify investment opportunities.
Intended Users	Finance students, retail investors, portfolio managers, and data analysts interested in Pakistan equity markets.
Tools Used	Python, Streamlit (web framework), Plotly (interactive charts), pandas, numpy, scipy





Conclusion

Lab #07 demonstrated the full data preprocessing pipeline from raw CSV ingestion through missing value analysis, duplicate detection, type correction, outlier handling, and feature engineering applied to a real-world financial dataset of 148,863 records. Seven visualisations covering histograms, boxplots, line charts, heatmaps, bar charts, scatter plots, and return heatmaps provided comprehensive exploratory coverage of the KSE-100 dataset.

Lab #08 built upon the cleaned data to deliver an eleven-chart interactive Streamlit dashboard with eight live KPI cards, sidebar filtering, and a full dark-mode UI. The dashboard enables non-technical stakeholders to explore five years of Pakistan stock market data through an intuitive, visually rich interface.

Together, both labs illustrate the end-to-end data science workflow: data quality assurance → statistical exploration → interactive communication of insights.